SEMESTER – V			
Course Name	**:** Full Stack Development Laboratory	**Course Code:**	BISFS508
No. of Lecture Hours / Week	**:** 00	**CIE Marks:**	50
No. of Tutorial / Practical Hours / Week	**:** 04	**SEE Marks:**	50
Total No. of Lecture + Tutorial / Practical Hours	**:** 48	**SEE Duration:**	02 Hrs.
L: T: P	**:** 0:0:4	**Credits:**	02

Course Prerequisites

Basic knowledge of programming in Javascript, Html and CSS

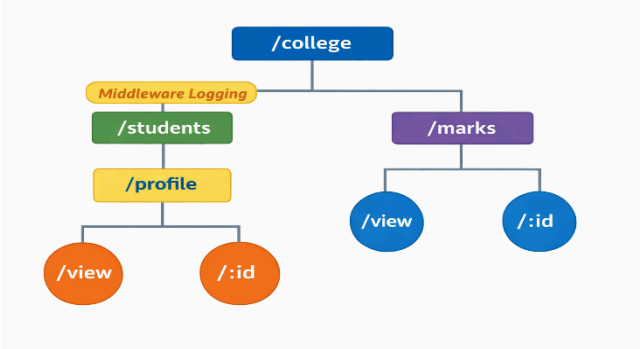
Course Overview

This course provides hands-on coding exercises on back-end programming on Express framework in javascipt

Course Learning Objectives (CLO)

This course will enable students to:

- Understand and use APIs of Express framework
- Develop backend components for web application
- Use node libraries for backend operations,

A- Demonstration 1 Using commands, setup a web application project using Express.js and run the sever to test its working. 2 Implement a simple web application using Express.js to demonstrate the working of middleware, controllers and Routers. 3 Implement a simple web application using Express.js to demonstrate error handling, static resources and rate-limiting mechanisms. ***B- Exercise*** 4 Design and configure hierarchical routing in Express.js using a modular approach with separate routes, enabling structured URL handling and applying middleware at appropriate route levels for efficient request processing and scalability. 	**48**

5 To design and implement a request validation mechanism in an Express.js application using express-validator by applying middleware functions to validate user input (such as name, email, and age)

6 Implement the POST and GET methods for a Student collection using Express.js and Mongoose to perform Create and Read operations. Return an appropriate message in the response.

7 Implement UPDATE and DELETE operations on a student collection using Express.js with Mongoose. Create suitable records in the collection to support the operations. Return an appropriate message as a response.

8 Implement an authentication API using Express.js, with passwords stored as one-way-encrypted hash values. Return an appropriate message as a response.

C- Structured Inquiry

9 To design and implement user authentication in an Express.js application using the Local Strategy of Passport.js, enabling secure verification of user credentials (username and password) through middleware-based authentication.

10 Design and implement role-based routing access in an Express.js application using middleware, enabling authorization of users based on their roles (such as admin and user) to control access to specific routes and ensure secure resource handling.

11 Implement traditional server-based session management using Express.js, wherein user session data is stored and maintained on the server to enable stateful authentication and secure tracking of user interactions across multiple HTTP requests.

12 To design and develop a secure session management system using JSON Web Tokens (JWT) in Express.js, enabling user authentication, token generation, and validation for protected routes without maintaining server-side session data.

Open Ended Experiments

The student can choose to solve any one open-ended problem to illustrate APIs for web applications in the domains specified below using Express and node packages.

1.E-Commerce

2.Food Delivery app

3. Management applications.

Course Outcomes (COs)

At the end of the course, students will be able to

CO1	Demonstrate the working of router, middleware and controller components of backend application
CO2	Implement REST API for a given requirements to interface with databases

CO3	Design and develop REST APIs for an application in team by exploring various programming tools and techniques

Mapping of CO – PO/PSO:

CO	PO											PSO			
	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PSO1	PSO2	PSO3	PS04
CO1	3												3		
CO2			3										3		
CO3					3			3	3		3		3		
Avg.	3		3		3			3	3		3		3		